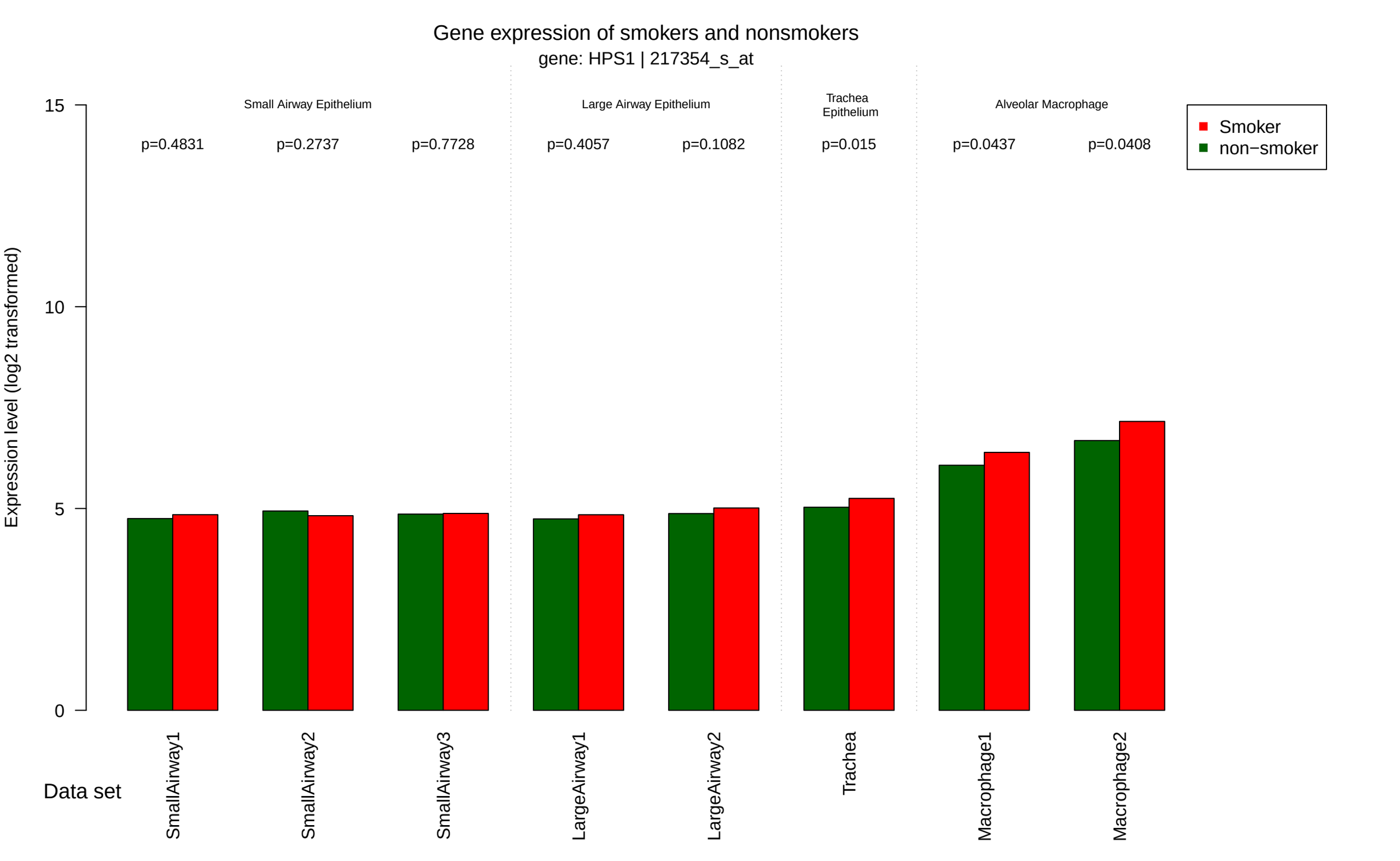
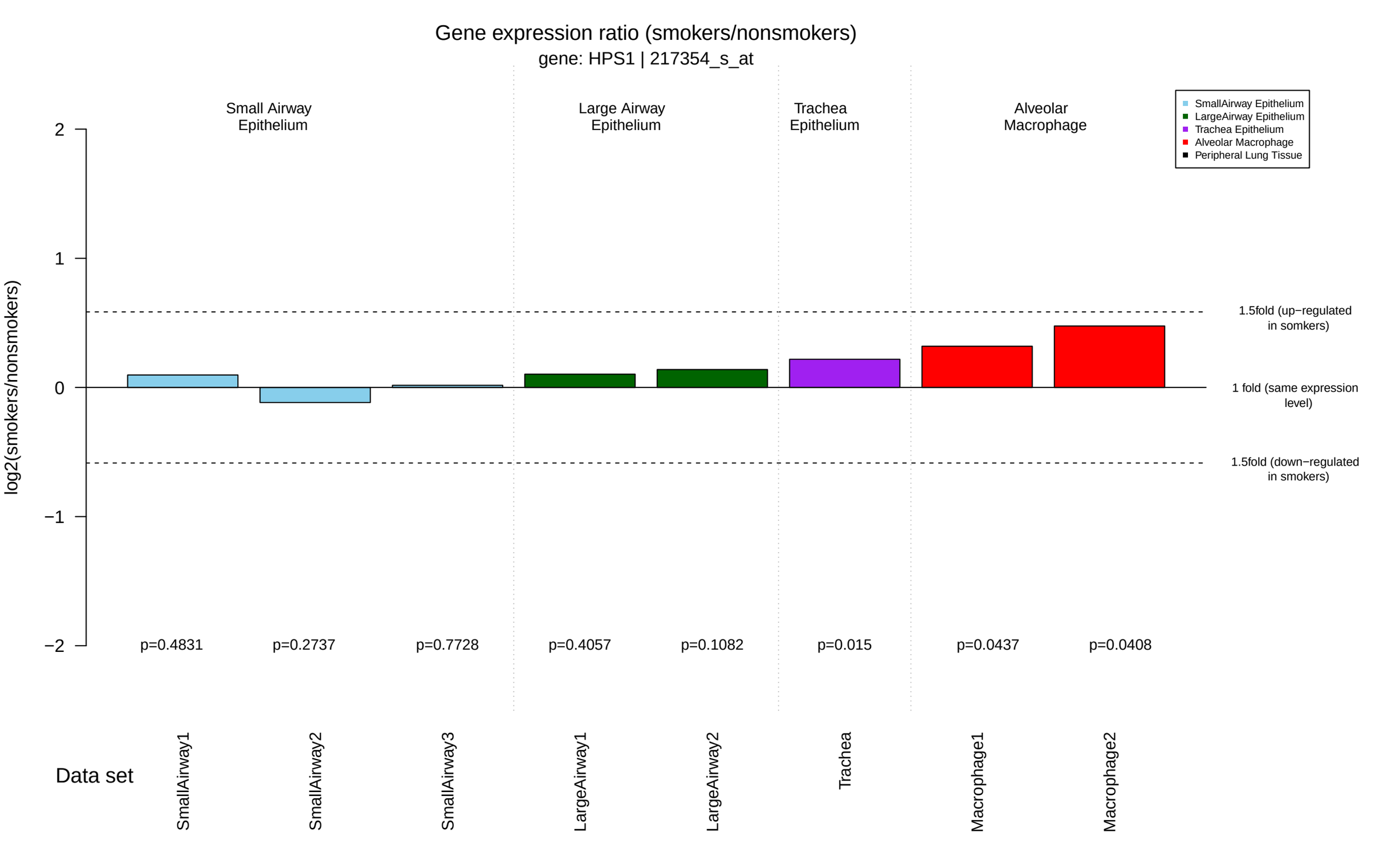
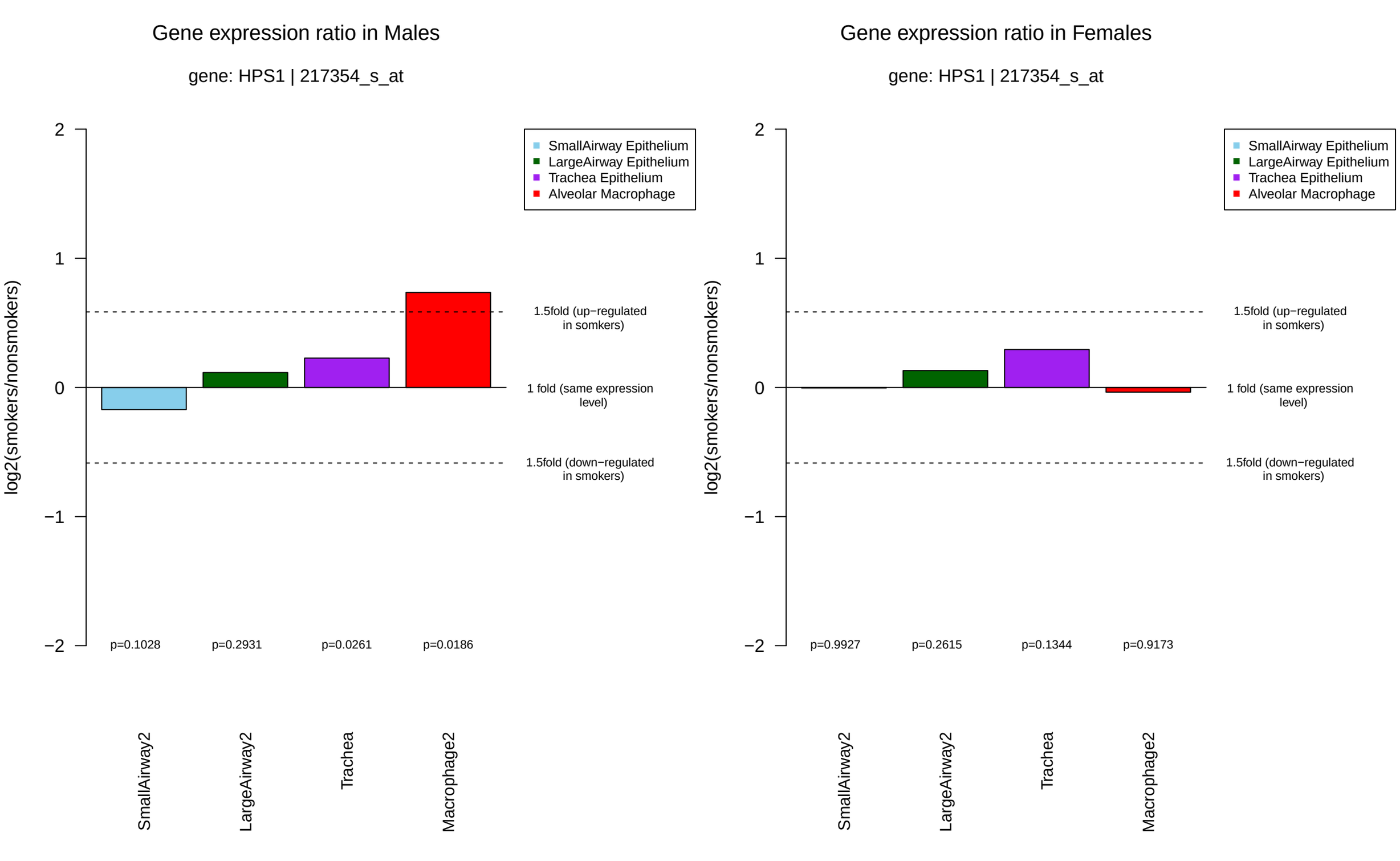
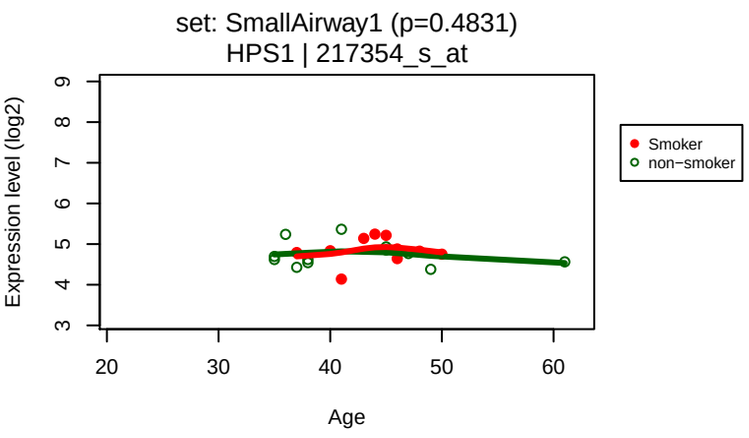
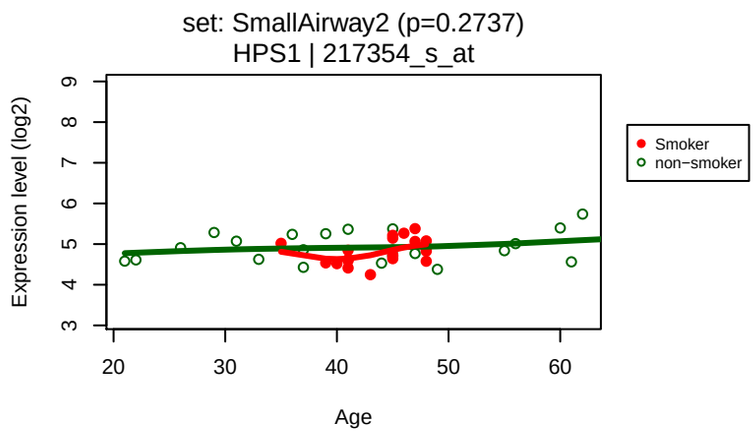
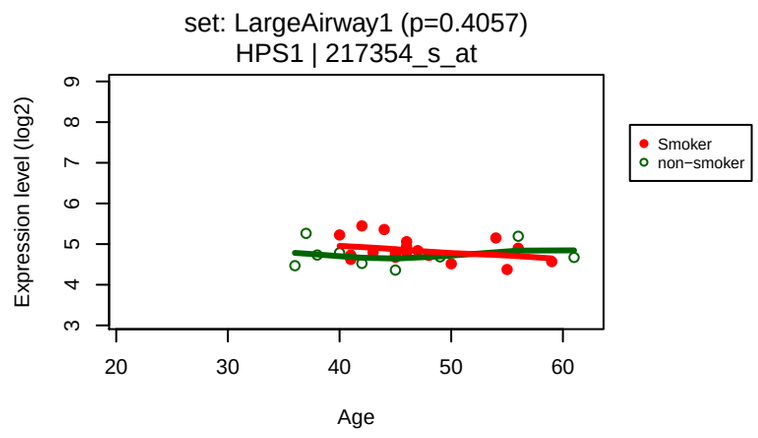

Gene expression of smokers and nonsmokers
gene: HPS1 | 217354_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4831
p=0.2737
p=0.7728
p=0.4057
p=0.1082
p=0.015
p=0.0437
p=0.0408
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

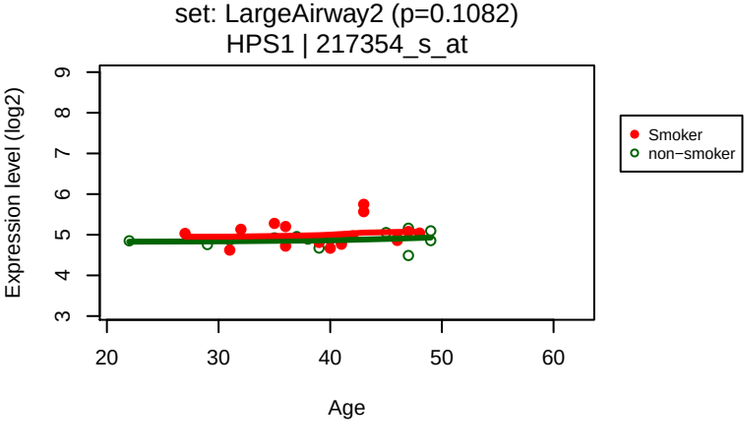

Gene expression ratio (smokers/nonsmokers)
gene: HPS1 | 217354_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4831
p=0.2737
p=0.7728
p=0.4057
p=0.1082
p=0.015
p=0.0437
p=0.0408
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

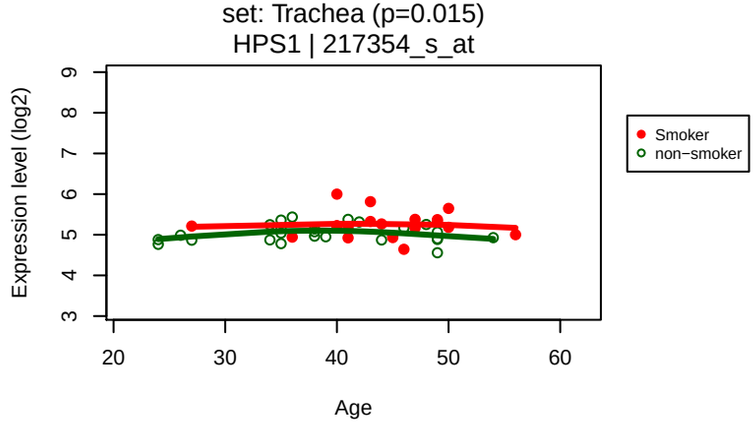

Gene expression ratio in Males
gene: HPS1 | 217354_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1028
p=0.2931
p=0.0261
p=0.0186
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HPS1 | 217354_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9927
p=0.2615
p=0.1344
p=0.9173
SmallAirway2
LargeAirway2
Trachea
Macrophage2

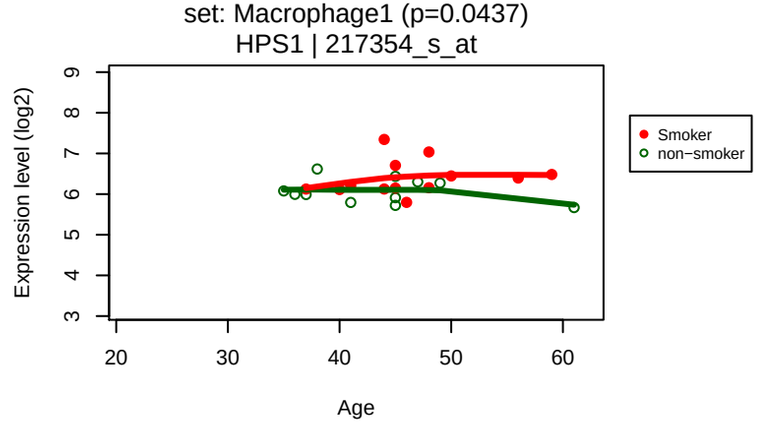

set: SmallAirway1 (p=0.4831)
HPS1 | 217354_s_at
Expression level (log2)
Age
Smoker
non−smoker

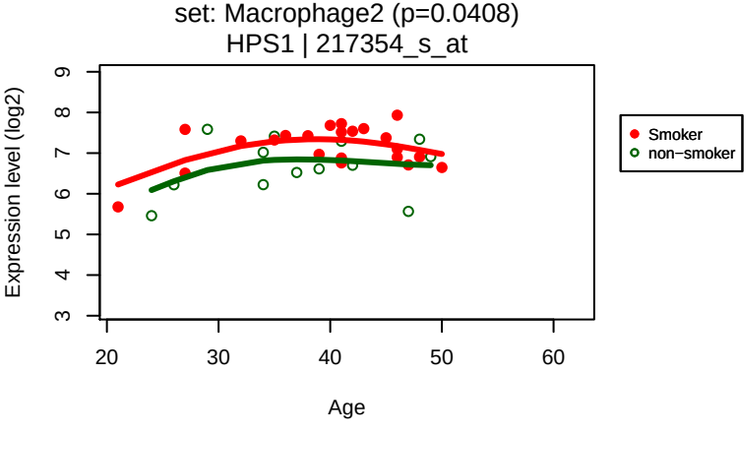

set: SmallAirway2 (p=0.2737)
HPS1 | 217354_s_at
Expression level (log2)
Age
Smoker
non−smoker

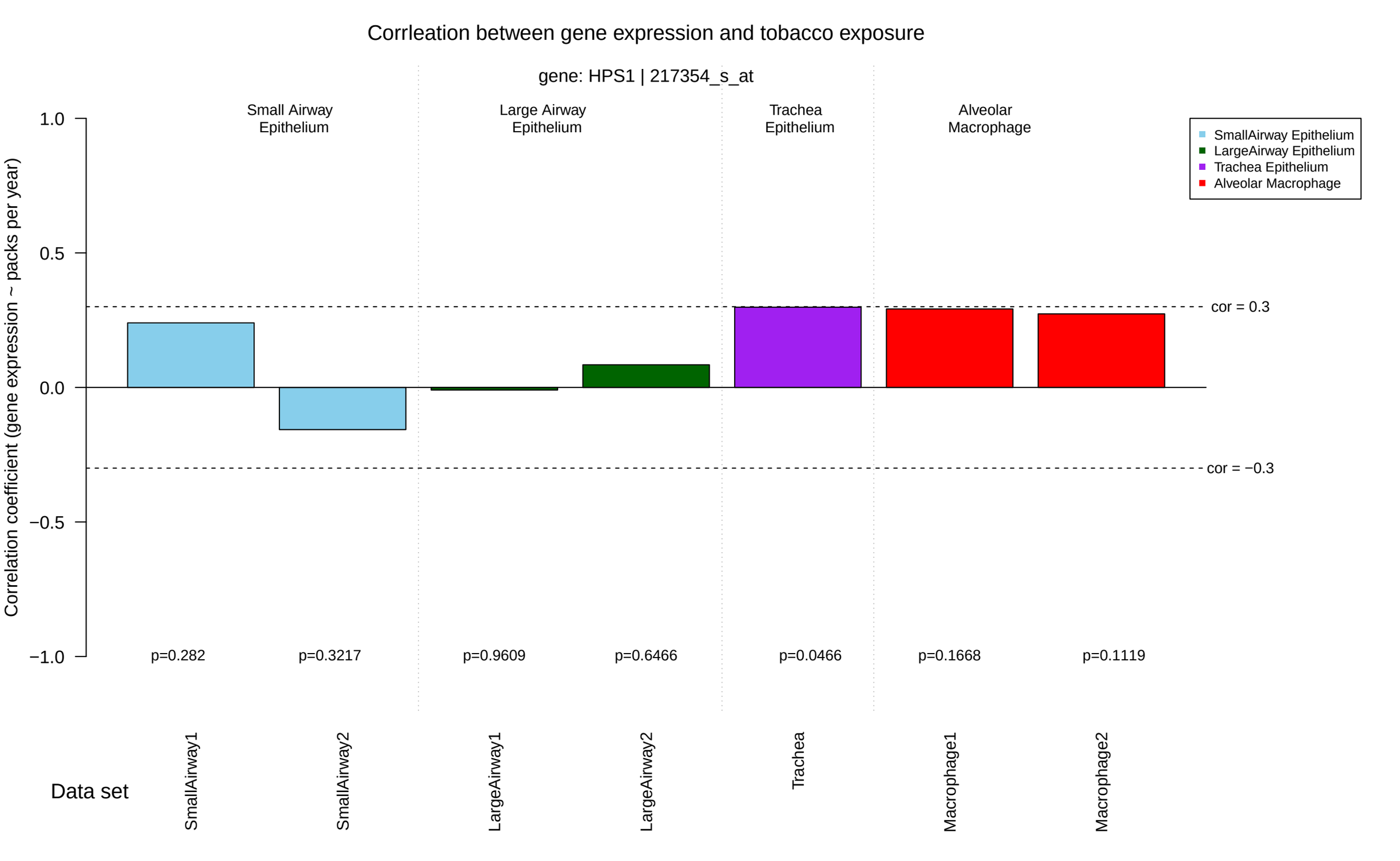

set: LargeAirway1 (p=0.4057)
HPS1 | 217354_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1082)
HPS1 | 217354_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.015)
HPS1 | 217354_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0437)
HPS1 | 217354_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0408)
HPS1 | 217354_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HPS1 | 217354_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.282
p=0.3217
p=0.9609
p=0.6466
p=0.0466
p=0.1668
p=0.1119
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2